Schweizerische Eidgenossenschaft
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Eidgenössisches Departement für Verteidigung, Bevölkerungsschutz und Sport VBS
Bundesamt für Rüstung - armasuisse ar

Research on e-ID Privacy and Security

Dr. Martin Burkhart
Scientific Project Manager

CYD
CYBER DEFENCE CAMPUS

Federal Council Requires an Unlinkable e-ID

The technical implementation of the e-ID and the necessary trust infrastructure will therefore take place in two stages. In the first stage, a highly secure trust infrastructure will be introduced. At the same time, work will be carried out on a solution that meets even higher privacy requirements. In particular, it should not be possible in future to attribute or trace uses of an e-ID to an individual. This quality should be realised as quickly as possible. To that end, CHF 1 million will be made available for research funding. Interested parties will be able to participate in the development of the solution.

news.admin.ch/en/nsb?id=102922
Bern, 6.12.2024

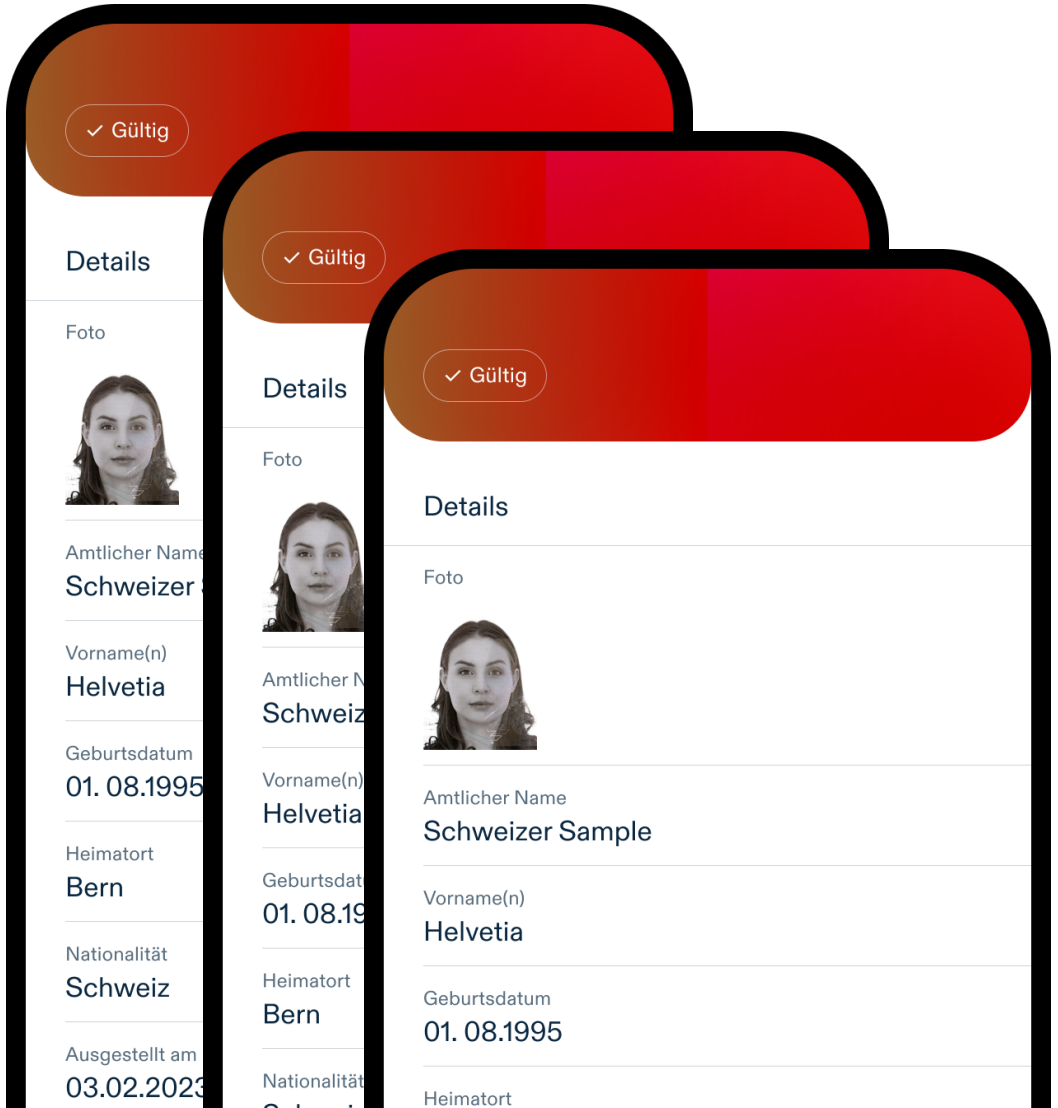
Unlinkability of e-ID in 2026 Through Batch-Issuance

Batch-Issuance

- Multiple key pairs issued for same VC
- Each key pair used only once
- Presentations unlinkable for verifiers

!

- Linkability through network metadata or issuer collusion still possible
- Issuer could learn usage statistics
- Scalability for large-scale adoption?



https://www.eid.admin.ch/en/so-wird-die-e-id-unverknuepfbar-e

Collaboration for Research Funding

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Collaboration with CYD Campus for research project management and funding

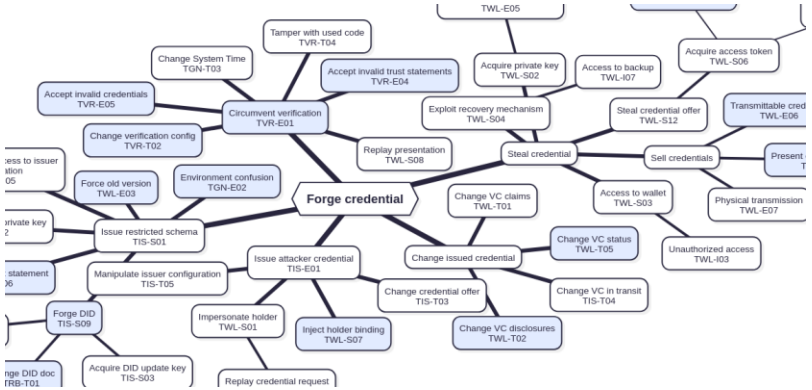
CYD
CYBER DEFENCE CAMPUS

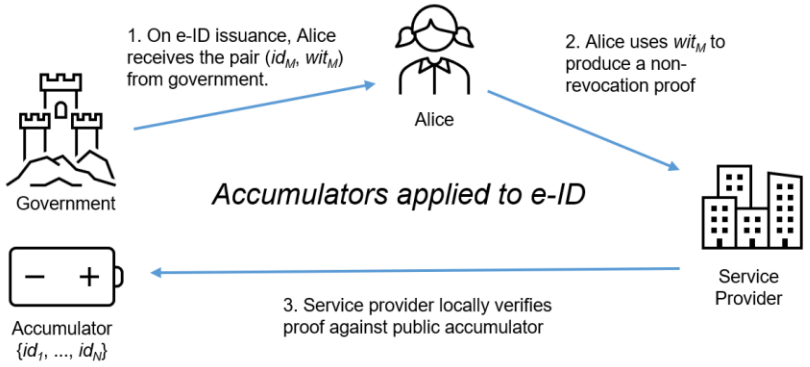
Why the Cyber-Defence Campus?

- Founded 2019, part of armasuisse Science & Technology
- Connecting government, industry and academia
- 60+ collaboration partners
- Strong ties with academia
  - Student theses supervision (~40 theses/year)
  - Commissioned research contracts
  - CYD Campus fellowships
- Collaboration on e-ID since 2023

Examples Research Projects in e-ID Context

- **Security Testing**
  - SWIYU infrastructure (presented here July 2025)
  - Mobile platform security
- **Privacy-Enhancing Technologies**
  - Zero-Knowledge Proofs
  - Credential Revocation
- **Distributed Key and Trust Management**
  - Social Key Recovery
  - KERI (Key Event Receipt Infrastructure)





https://github.com/cyber-defence-campus/self-sovereign-identity

Invited Research Contributions

- **Scope**
  - unlinkability
    - zero-knowledge proofs (ZKP)
    - credential revocation
    - unobservability
  - post-quantum security
  - accountability / non-repudiation
  - trust management
- **Minimal Requirements**
  - Relevant to e-ID project
  - No (good) technical solution available
  - Scientific contribution

How Does the Funding Work?

Research Proposal

- Submission website
- Two-stage process
- Expert jury + mentor

E-ID Research Committee
- 2x BJ
- 1x BIT (FOITT)
- 1x CYD Campus

Relevance for e-ID?

CYD Fellowships for
- Post Docs
- PhD students
- Master students

EPFL CYD CYBER DEFENCE CAMPUS

Commissioned Research
- Internal research programs
- Researchers approached directly

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Contracts

First Project: Quantum-Resilient Swiss e-ID

- Prof. Dr. Alexandre Duc @ HEIG-VD / HES-SO
- Time: Feb. 2026 – Dec. 2026
- Scope of project:
  - E-ID stack analysis
  - Impact analysis based on scenarios
  - Implications of PQC on e-ID
  - Impact on batch-issuance
  - PoC and demo of quantum-safe e-ID stack
  - Overview on post-quantum secure enclaves

Thanks for your attention

CYD Fellowships:

Contact:
cydcampus@ar.admin.ch

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